

TISSUE RUN UP PROTOCOL

1. Place perfusion fixed tissue into 2.5% glutaraldehyde in 1x PBS (pH 7.4) for 120+ minutes.
2. Cut up tissue to the proper size keeping in mind any orientation needed in the final block.
3. Wash in 3 changes 1x PBS for 15 minutes each.
4. Post-fix specimens in for 1 hour in 1% OsO₄ with 1% potassium ferricyanide.
5. Wash in 3 changes 1x PBS for 15 minutes each.
6. Dehydrate in graded series of alcohol for 15 minutes each:
 - a. 30% ethanol
 - b. 50% ethanol
 - c. 70% ethanol
 - d. 90% ethanol
 - e. 100% ethanol x 3
7. Dehydrate further in two 10 minute changes of propylene oxide. *
8. Infiltrate with a 1:2 mix of propylene oxide and epon for 3+ hours followed by 1:1 mix of propylene oxide and epon overnight.
9. Infiltrate with pure epon overnight at 4° C or 1 hour at room temperature.
10. Continue infiltration with three 1 hour changes of epon (can be done under vacuum).
11. Embed in pure epon at 37° C for 24 hours.
12. Cure for 48 hours at 60° C.

Blocks are now ready to section, stain, and scope.